

Tanmay Kulkarni

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Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, Kotlin, Bash

Databases: MongoDB, NoSQL, MySQL, SQLite3, SQL, PostgreSQL, DynamoDB

Web/Cloud: NodeJS, SpringBoot, Linux (Debian, Arch), AWS SageMaker, AWS S3, Express.JS, Flask, React, Angular, Next.JS

Tools: Docker, Kubernetes, Git, Matplotlib, NumPy, Pandas, NLTK, AndroidSDK, CMake, OpenGL, TensorFlow, PyTorch, Maven

Certifications:

- **Coursera:** Machine Learning Specialization, Deep Learning Specialization
- **Udemy:** Web Development Bootcamp
- **AWS Academy:** Machine Learning Foundations, Cloud Foundations

Experience

All India Council for Technical Education, India

AI-ML Intern

May 2023 – Jul 2023

- Deployed three production SageMaker endpoints with average uptime over 99% and average response time of less than 200ms
- Reduced data preprocessing time by over 30% by building scalable Glue ETL jobs and optimized S3 partitioning
- Achieved a cost reduction of over 20% on ML training workloads using spot instances and optimized instance selection
- Automated labeling pipeline with SageMaker Ground Truth, achieving 92% label quality and processing 20k items in 2 weeks

YBI Foundation, India

Machine Learning Intern

Feb 2023 – May 2023

- Constructed regression models in SciKit-Learn for an internal dataset of over 10k entries, reducing prediction error by over 15%
- Identified five key feature correlations using regression and decision-tree analysis, informing strategic improvements
- Created and showcased 10+ interactive dashboards to visualize model performance, highlighting areas needing improvement
- Improved the top model's accuracy from 64% to 95% through hyperparameter tuning and feature engineering

Projects

FusionSage – Web App for Retrieval Augmented Generation System with Multi-Source Data Integration

Technologies: ChatGPT API, LangChain, SQLite3, Pinecone, CassandraDB, BeautifulSoup4, Python

- Produced a full-stack RAG pipeline using LangChain and ChatGPT API to aggregate information from web, PDF, and database sources, increasing information retrieval accuracy by over 45%
- Scaled by integrating Pinecone indexes and horizontally scaling CassandraDB, supporting over 200 concurrent test users
- Processed and chunked content from 1k documents using BeautifulSoup4 into 200 word chunks for precise information retrieval
- Incorporated LangChain to integrate the GPT 3.5 turbo model for generating answers based on the searched text chunks
- Employed the SQL utility from LangChain to process the database script into a SQLite3 database for GPT 3.5 turbo

CareerSpeak – Web App for Resume-Based Job Matching with Speech Interviews and Grammar Analysis

Technologies: React Native, Docker, Python, Flask, Node.JS, ExpressJS, MySQL, REST API

- Led a team of 4 to develop a cross-platform job-matching app (web, Android, iOS) that offers resume grammar checks, voice interview practice, and job recommendations
- Architected and deployed a fault-tolerant microservices backend (10 Docker containers) using Node.js (Express.js) and Python (Flask), supporting scalable user authentication, job matching, and data processing
- Integrated NLP components (LanguageTool, transformer-based grammar models, WhisperAI) to achieve over 95% accuracy in grammar correction and paraphrasing
- Prepared REST APIs linking the React Native front end to a MySQL backend, achieving under 200ms latency under simulated high traffic loads for a smooth UX

WebPlots – Plot Generator for Python and R Scripts

Technologies: Angular, Python, R, Matplotlib, Ggplot2, PlotlyJS, Flask, Docker

- Built an Angular web application executing user-provided Python and R scripts to render plot images for data visualization in under 200ms
- Integrated support for Matplotlib, Plotly, and ggplot2 libraries, allowing users to generate 2D and 3D interactive plots
- Designed a secure, containerized Flask backend with input validation and sandbox for safe visualization workflows

Education

Master of Science in Computer Science

Indiana University Bloomington

Aug 2024 – May 2026

GPA: 3.52

Relevant coursework: Software Engineering II, Applied Algorithms, Data Visualization, Data Mining

Bachelor of Engineering in Artificial Intelligence and Machine Learning

Savitribai Phule Pune University

Aug 2020 – May 2024

GPA: 3.46

Relevant coursework: Web Technologies, Cloud Computing, Distributed Systems, Deep Learning